



Running the Right Script in Autopilot

Here is a step-by-step guide to running a programming script effectively:

1. Selecting the Scripting Environment:

- **Identify Supported Languages:** Determine the scripting languages and environments compatible with your autopilot system. For instance, ArduPilot supports Lua and Python, while PX4 uses NuShell (NSH) or MAVLink shell (MAVSH) commands.

2. Preparing the Script:

- **Develop or Obtain the Script:** Create or acquire a programming script that provides the desired functionality for the drone autopilot system. Ensure that the script follows the syntax and guidelines specific to the chosen scripting environment.

3. Connecting to the Autopilot System:

- **Establish Connection:** Use the appropriate ground control station (GCS) software or command interface to connect to the drone autopilot system. Examples include Mission Planner for ArduPilot or QGroundControl for PX4.
- **Connect via Telemetry or Serial:** Connect to the autopilot system using a telemetry link (e.g., radio, WiFi, USB) or a direct serial connection, depending on the hardware setup and communication protocol supported.

4. Accessing the Scripting Interface:

- **Navigate to the Scripting Interface:** Access the scripting interface or console within the GCS software or command interface. Follow the instructions provided in the autopilot system documentation to reach the scripting functionality.



5. Uploading or Executing the Script:

- **Upload the Script:** If required, upload the programming script to the autopilot system, adhering to the instructions provided in the system documentation and scripting interface.
- **Direct Execution:** Alternatively, if the system allows, copy and paste the script code into the scripting interface for direct execution.

6. Verifying Script Execution:

- **Check for Errors:** Confirm that the script has been successfully uploaded or executed by checking for any error messages or feedback from the scripting interface.
- **Monitor System Behaviour:** Observe the behaviour of the autopilot system and the drone during script execution to ensure that the intended functionality is achieved, and the script operates as expected.

7. Testing and Validation:

- **Conduct Testing:** Test the functionality implemented by the script in a controlled environment, such as a simulated flight or a testbed, to validate its performance and integration with the autopilot system.
- **Address Issues:** Perform thorough testing to identify and resolve any issues, errors, or unexpected behaviours during script execution.

8. Documentation and Reporting:

- **Document the Process:** Record the procedure for running the programming script in the autopilot system, including configuration settings, dependencies, and prerequisites for successful execution.
- **Prepare Reports:** Create documentation and reports summarising the results of the script execution, including observed behaviours, outcomes, and recommendations for further actions or improvements.